Melissa Ann Collier, Ph.D.

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EDUCATION

2018- 2023 Ph.D. in Biology, Georgetown University, Washington, DC,
"The impact of behavior on pathogen transmission: a data-driven mathematical modeling approach."
Dissertation Defense Passed with Distinctions

2016-2017 M.P.S, Marine Mammal Science, University of Miami, Miami, FL,
"Quality assessment and quality control (QA/QC) of the Biscayne Bay bottlenose dolphin photo-identification project database and spatial plotting of track-lines."
Masters Thesis

2011-2015 B.S., Biochemistry, Florida Southern College, Lakeland, FL
"A diet analysis of resident sharks in Tampa Bay using DNA barcoding for prey identification."
Senior Honors Thesis.

PUBLICATIONS

Preprints

Collier, M.A., Urian, K., Theisen, S., Jacoby, A-M, Wilkin, S., Patterson, E.M., Wallen, M., Colizza, V., Mann, J., Bansal, S. (2024) Seasonal contact and migration structure mass epidemics and inform outbreak preparedness in bottlenose dolphins. *Accepted.* Preprint on biorXiv: https://doi.org/10.1101/2024.10.02.616317

Albery, G.F., Becker, D.J., Firth, J.A., Silk, M., Sweeny, A.R., Eric Vander Wal, E., Webber, Q.,…**Collier, M.A**.,…Bansal., S. Density-dependent network structuring within and across wild animal systems. (2024). *In Revision*. Preprint on biorXiv: https://doi.org/10.1101/2024.06.28.601262

Journal Articles

Collier, M.A.*, Jacoby, AM., Foroughirad, V., Patterson, E.M., Krzyszczyk, E., Wallen, M.M., Miketa, M.L., Karniski, C., Wilkin, S., Mann, J., Bansal, S. Breathing synchrony shapes respiratory disease risk in bottlenose dolphins. *Commun Biol* 8, 870 (2025). https://doi.org/10.1038/s42003-025-08161-1

Murphy, C.J., **Collier, M.A.*,** Jacoby, A-M., Patterson, E.M., Wallen, M.M., Mann, J., Bansal, S. Automated skin lesion detection and prevalence estimation in Tamanend's bottlenose dolphins. (2025) *Ecological Informatics* 85:102942. https://doi.org/10.1016/j.ecoinf.2024.102942
*Corresponding author
Rippel, T.M., DeCandia, A.L., **Collier, M.A.,** McIntosh, C.L., Murphy, S.M., and Wimp, G.M. Habitat characteristics and plant community dynamics impact the diversity, composition, and co-

occurrence of sediment fungal communities in salt marshes. (2023) *Wetlands* 44(3):2024. https://doi.org/10.1007/s13157-023-01756-6

MacLaren NG, Meng L, **Collier M.A**., and Masuda N. (2024) Cooperation and the social brain hypothesis in primate social networks. *Frontiers in Complex Systems*. 1:1344094. https://doi.org/10.3389/fcpxs.2023.1344094

Dolezal, M., Foroughirad, V., Fish, F., Murphy, C., **Collier, M.A**, Jacoby, AM., Rittmaster, K., Bansal, S., Mann, J. (2023) Some like it hot in the flow: Physiological, environmental, and hydrodynamic factors in *Xenobalanus globicipitis* attachment to cetaceans. *Marine Mammal Science* 39(3):961-975. https://doi.org/10.1111/mms.13022

Collier M.A, Albery G.F., McDonald G.C. and Bansal S. (2022) Pathogen transmission modes determine contact network structure, altering other pathogen characteristics. *Proceedings of the Royal Society B*. 289(1989):20221389. https://doi.org/10.1098/rspb.2022.1389

Book Chapters

Collier, M.A, Mann, J., Ali, S., Bansal, S. (2022). Impacts of Human Disturbance in Marine Mammals: Do Behavioral Changes Translate to Disease Consequences?. In: Notarbartolo di Sciara, G., Würsig, B. (eds) Marine Mammals: The Evolving Human Factor. Ethology and Behavioral Ecology of Marine Mammals. Springer, Cham. https://doi.org/10.1007/978-3-030-98100-6_9

CONFERENCE PRESENTATIONS

Collier, M., Urian, K., Jacoby, A-M, Mann, J. Bansal, S. Seasonal contact and migration structure mass epidemics and inform outbreak preparedness in bottlenose dolphins. Biennial Conference on the Biology of Marine Mammals. Oral Presentation November 13, 2024.

Collier, M., Mann, J., Jacoby, AM, Foroughirad, V., Bansal, S. Characterizing infectious disease risk in a sentinel marine mammal with social behavior data. Animal Behavior Society Annual Meeting. Oral Presentation. July 12th, 2023.

Collier, M. Mann, J., Bansal, S. Modeling the impacts of human disturbance in cetaceans and pinnipeds: Do behavioral changes translate to disease consequences? Society of Marine Mammalogy Conference. Oral Presentation. August 1-5th, 2022.

Collier, M. Mann, J., Bansal, S. Modeling the impacts of human disturbance in cetaceans and pinnipeds: Do behavioral changes translate to disease consequences? SEAMMAMS. Oral Presentation. May 20th, 2022.
Awards: *Best Graduate Student Presentation*

Collier, M. Albery, G., Sah, P., McDonald, G., Pizzari, T., Bansal, S. Pathogen's transmission modes determine contact network structures, altering other pathogen characteristics. Ecology and Evolution of Infectious Disease. Poster. June 14th- 17th, 2021

FELLOWSHIPS, GRANTS, AND AWARDS

2024 PI Morris Animal Foundation Fellowship Training Award D24ZO-425:

	A comprehensive study of disease transmission dynamics in a wild marine mammal: seasonality, population fluctuations, and climate influences ($144,805)
2022	Exploration and Field Research Grant. ECWG ($3900)
2022	Grad Gov Research Project Award. Georgetown University. ($500)
2022	Best Graduate Student Oral Presentation. SEAMAMMS Conference.
2021	Edward O. Wilson Conservation Award. Animal Behavior Society ($1500)
2015	FSC Department of Biology Poster Competition, Third Place ($250)
2013-2015	NOAA Ernest F. Hollings Scholarship ($8000/year plus paid summer internship, $650/week)
2011-2015	FSC Honors Program Scholarship ($1500/year)

PROFESSIONAL EXPERIENCE

2021-2022	Field Team Leader, Potomac Chesapeake Dolphin Project, Washington DC
2018-2020	Field Assistant, Potomac Chesapeake Dolphin Project, Washington DC
2018-Present	Data Analyst, Potomac Chesapeake Dolphin Project, Washington DC.
2018-Present	Data Manager, Animal Social Network Repository, Washington DC.
2017	Marine Endangered Species Observer, REMSA INC.
2016-2017	Bottlenose Dolphin Photo ID Intern, NOAA
2015-2016	Marine Environmental Educator, Driftwood Education Center

PUBLIC PRESENTATIONS, SERVICE AND OUTREACH

2025	Peer Reviewer: New Jersey Sea Grant Consortium Offshore Wind Research Proposals
2023-Present	Ad hoc Peer Reviewer: *Royal Society Open Science* *Ecology Letters* *Behavioral Ecology and Sociobiology*
2024	Northern Neck Soil and Water Conservation Society Round Table
2021	Potomac Conservancy Riverside Chat
2020-2022	Secretary, Biology Organization of Graduate Students
2020	Graduate Student Union Bargaining Committee Member
2017	Dolphin Anatomy and Echolocation. Frost Museum of Science Miami, FL.
2017	The Dolphins of Biscayne Bay. Miami Science Barge. Miami, FL.
2017-2018	Marine Animal Rescue Society (MARS). Volunteer Responder
2017	University of Miami, Avian and Wildlife Laboratory, Department of Clinical Pathology. Volunteer Technician.

TEACHING

2021-2022	Guest Lecturer, Bio 019, "Marine Mammals in the DMV", Georgetown University
2020	Teaching Fellow, BIO 422, Modeling Biological Populations, Georgetown University
2019	Teaching Fellow, BIO 423, Networks in Biology, Georgetown University
2017	Guest Lecturer, MBE 604, "Introduction to Dorsal Fin Matching Using Finbase 2.0", University of Miami.
2017	Guest Lecturer, MSC 350, "Introduction to Cetacean Photo-Identification Studies and Finbase 2.0", University of Miami.
2015-2016	Environmental Educator. Driftwood Education Center, St. Simon's Island, GA

RESEARCH FEATURED IN PRESS

Collier, M.A. November, 2023. "River of Intrigue: The Potomac's Dolphins and Their Disease Saga" *Potomac Conservancy Blog.* https://potomac.org/blog/2023/11/1-dolphin-diseases-updates

Larson, C. 7 April, 2022. "Dolphins' playful social habits form bonds, but spread virus." *AP NEWS*. https://apnews.com/article/dolphins-social-habits-form-bonds-spread-virus-2602b5333d880c57060fdbe917989516

Pipkin, W. 13 June, 2022. "Chesapeake dolphins thrill spotters, scientists." *Bay Journal.* https://www.bayjournal.com/news/wildlife_habitat/chesapeake-dolphins-thrill-spotters-scientists/article_be6594be-eb04-11ec-a371-0f9f133667ef.html

Jacoby, AM. 13 October, 2021. "Potomac River dolphins also suffer from viral outbreaks- here's how to help." *Potomac Conservancy Blog*. https://potomac.org/blog/2021/10/4/dolphin-disease-blog

Cuomo, J. 18 October 2019. "Dolphin population growing in Potomac River." *Georgetown Voice.* https://georgetownvoice.com/2019/10/18/dolphin-population-growing-in-potomac-river/